

# Hackathon Problem Statements

Regd No: 25091A32D5, 25091A3281

Problem Statement No: 7

**Rural Supply Chain Route Optimizer** Design a farm-to-warehouse logistics optimizer using Dijkstra's or A\* algorithm to find the shortest, least-cost delivery routes for perishable produce across rural road networks. Incorporate real or simulated traffic and road-quality data to adjust route weights dynamically, and use regression to predict spoilage risk based on transit time and temperature. Model trucks, warehouses, and farms as OOP entities with capacity constraints. The objective is reducing the 20-30% post-harvest loss common in Indian agri-supply chains.

Regd No: 25091A32B3, 25091A3282

Problem Statement No: 40

**Smart Streetlight Energy Optimizer** Create an energy optimization system for streetlights that uses clustering on simulated occupancy / motion sensor data to identify low-traffic zones and time periods where dimming is safe, reducing energy consumption. Apply time-series analysis to correlate lighting needs with seasonal daylight variation. Use OOP to model streetlight units with adjustable brightness states. This can significantly cut municipal electricity costs while maintaining public safety standards.

Regd No: 25091A32F6, 25091A32G3

Problem Statement No: 53

**City Budget Allocation Simulator** Design a municipal budget allocation simulator using dynamic programming (a knapsack variant) to optimally distribute a fixed annual budget across competing needs like roads, sanitation, healthcare, and education based on impact scores. Use OOP to model budget categories, projects, and constraints. This tool can help city councils visualize trade-offs and make more transparent, data-informed budget decisions.

Regd No: 25091A3288, 25091A32N7

Problem Statement No: 69

**Career Path Recommendation Engine** Design a career recommendation system using decision tree or random forest models trained on student interests, aptitude scores, and academic strengths to suggest suitable career paths and higher education options. Use OOP to model career categories, required skills, and student profiles. This provides much-needed structured career guidance often unavailable in government and rural schools.

Regd No: 25091A3233, 25091A3234

Problem Statement No: 75

**River Water Quality Classifier** Build a water quality classification system using machine learning models trained on chemical parameters (BOD, DO, pH, coliform count) from public river monitoring datasets to categorize water quality into usability grades. Use OOP to model monitoring stations, samples, and classification results. This supports pollution control boards in prioritizing river stretches needing urgent remediation.

Regd No: 25091A32B8, 25091A32C0

Problem Statement No: 101

**Cryptocurrency Price Volatility Analyzer** Build a volatility analysis tool using statistical time-series methods (standard deviation, rolling volatility) on public cryptocurrency price datasets to help users understand risk levels before investing. Visualize volatility trends alongside price movements. Use OOP to model different cryptocurrencies and analysis metrics. This promotes informed, risk-aware financial decision-making among young, increasingly crypto-curious investors.